Ziad Mahmoud Hanafi

Software Engineer | Machine Learning Engineer
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Summary

Final-year Computer Engineering student at E-JUST with hands-on experience in machine learning, computer vision, and data analysis. Currently enrolled in the **DEPI AWS ML Engineer Track** and trained under Andrew Ng's ML curriculum. Interned at the Ministry of Communications and Information Technology (MCIT) within a national digital transformation initiative. Proven ability to build end-to-end ML pipelines, deploy scalable systems, and deliver data-driven insights using modern AI frameworks. Seeking an AI internship within a government-aligned institution to apply technical depth toward impactful, real-world outcomes.

Education

Egypt-Japan University of Science and Technology (E-JUST) — **New Cairo, Egypt**
Bachelor of Science in Computer Engineering *GPA: 3.2 / 4.0* — *Sep 2023 – Jun 2026*

- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, Cryptography, Database Systems, Artificial Intelligence, Computer Vision, Data Engineering, Secure Systems

October STEM High School — **Cairo, Egypt**
STEM Diploma — Valedictorian *GPA: 4.0 / 4.0* — *Sep 2019 – Jun 2023*

Experience

Ministry of Communications and Information Technology (MCIT) — **Cairo, Egypt**
AI & Data Analysis Intern — *Jul 2025*

- Worked within a national digital transformation initiative, conducting data analysis using **SPSS** and **Power BI** and delivering interactive dashboards for non-technical government stakeholders.
- Built data pipelines and **Power BI** reporting workflows that translated raw operational datasets into actionable business intelligence for government decision-making.
- Developed backend modules in **C# and .NET**, applying OOP principles to build reliable internal data-processing tools.
- Applied **AI ethics** frameworks — covering fairness, accountability, and transparency — in the context of responsible public sector technology deployment.

Alignerr (Labelbox) — **Remote**
AI Trainer & Model Evaluator — *2025 – Present*

- Contributed to the alignment of frontier generative AI models on **Labelbox's** selective AI training platform, evaluating model outputs against logical, ethical, and factual quality frameworks.
- Authored structured **prompts, instructions, and ranked responses** across ML and software engineering domains, producing high-quality RLHF training data for generative AI fine-tuning pipelines.
- Maintained top-tier accuracy scores across task batches, qualifying for advanced project access on the platform's performance-based ranking system.

Digital Egypt Pioneers Initiative (DEPI) — AWS ML Engineer Track — **Cairo, Egypt**
Trainee — Machine Learning Engineering on AWS, Round 5 — *2026 – Present*

- Selected for Egypt's government-backed national AI upskilling program, gaining hands-on experience with **AWS SageMaker**, cloud ML pipelines, and MLOps workflows.
- Covering production ML engineering topics including feature engineering, model evaluation, scalable data pipelines, and cloud deployment within a structured, accredited curriculum.

Empire Robotics & Innovation Hub | STEM Technology Club — **Cairo, Egypt**
Co-Founder & Technical Director | Workshop Leader — *2019 – 2023*

- Co-founded a student tech incubator and mentored **20+ peers** in robotics and AI; led weekly workshops for 50+ students, growing club membership by **150%**.

Projects

InkVision OCR System | *PyTorch, OpenCV, REST API* **2024**

- Built an OCR system using **PyTorch** with custom attention mechanisms, achieving **95% accuracy** across multi-language text recognition including Arabic.
- Deployed a scalable **RESTful API** backend serving the model to a mobile frontend with low-latency, high-availability cloud endpoints.

Tomato Disease Classification Pipeline | *PyTorch, TensorFlow, Computer Vision* **2024**

- Designed a complete end-to-end **ML pipeline** — data ingestion, augmentation, class-imbalance handling, model training, and edge-optimized deployment — for multi-class plant disease classification.
- Optimized deep learning architectures for **resource-constrained edge environments**, reducing inference latency while preserving classification accuracy.

MoneyFlow & IllumiScan | *Flutter, Node.js, MongoDB, OpenCV* **2023 – 2024**

- **MoneyFlow:** Full-stack expense management app built with clean architecture across Flutter frontend and Node.js/MongoDB backend, with real-time analytics and secure authentication.
- **IllumiScan:** Automated defect detection system using **OpenCV**, achieving **92% accuracy** in lab conditions through custom image preprocessing and contour-based detection pipelines.

Technical Skills

AI & Machine Learning: PyTorch, TensorFlow, OpenCV, Scikit-learn, Deep Learning, Computer Vision, NLP, ML Pipelines, Responsible AI / AI Ethics
Data Analysis & Visualization: Power BI, SPSS, Pandas, NumPy, Matplotlib, Data Wrangling, Dashboard Design
Backend & Web: Node.js, Express.js, C# (.NET), REST API Design, TypeScript, HTML/CSS
Mobile & Databases: Flutter, Dart, React Native, MongoDB, PostgreSQL, Firebase
Programming Languages: Python, C++, TypeScript/JavaScript, C#, Dart, Java
Cloud & MLOps: AWS (SageMaker, S3, EC2), Docker, Git, GitHub, CI/CD, Postman, Agile/Scrum
Software Engineering: Data Structures, OOP, Design Patterns, Clean Architecture, Algorithms

Languages

Arabic — Native proficiency | **English** — Full professional proficiency | **Japanese** — Intermediate (conversational)

Certifications & Honors

Certifications & Courses:

Machine Learning Specialization — Andrew Ng / DeepLearning.AI (Coursera)
Deep Learning Specialization — Andrew Ng / DeepLearning.AI (Coursera)
DEPI AWS ML Engineer Track — Digital Egypt Pioneers Initiative (In Progress)
Digital Transformation & AI Specialization — Stanford University
Industry 4.0 & Robotics — University of Pennsylvania
AI Ethics: Fairness, Accountability & Transparency in Machine Learning (Coursera)
Responsible AI Development Practices — Government Digital Academy, MCIT
Data Visualization with Power BI — Microsoft Learn

Awards & Competitions:

Winner — NASA Space Apps Challenge (2021)
National Finalist — Intel ISEF Egypt (2021)
3rd Place — IEEE Micromouse Competition (2019)